

High-performance tangential flow filtration: a highly selective membrane separation process

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Abstract

High-performance tangential flow filtration (HPTFF) is a new technology for protein and nucleotide purification. Conventional ultrafiltration (UF) is limited to separation of solutes that differ by tenfold in size. HPTFF is a two-dimensional purification method that exploits differences in both size and charge characteristics of biomolecules. It is hence possible to separate biomolecules with the same molecular weight. It is even possible to retain one biomolecule while passing a larger molecular weight species through the membrane. Current processes often use ion-exchange chromatography, UF and size exclusion chromatography (SEC) in three separate steps for purification, concentration and buffer exchange. HPTFF makes it possible to perform all of these steps in a single-unit operation, thereby reducing production costs. HPTFF uses the same linear scale-up principles already established for UF. Since HPTFF builds on existing UF technology, there is already a well-established industrial infrastructure in place for implementation of HPTFF processes. HPTFF can provide high-resolution purification while maintaining the inherent high throughput and high yield characteristics of conventional UF. HPTFF can therefore be used in initial, intermediate, and final purification stages. This presentation overviews the key aspects of this technology in terms of operation and optimization and reviews several real-world separations in biotech applications such as IgG and monoclonal antibodies.

Keywords: Ultrafiltration; Process compression; Void-free membranes; Protein purification

1. Introduction

High-performance tangential flow filtration (HPTFF) is a new protein purification technology.

Conventional tangential flow filtration is limited to the separation of solutes that differ by tenfold in size (e.g., cell–protein, virus–protein and protein–buffer). HPTFF is a two-dimensional purification method that exploits differences in

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both size and charge characteristics of biomolecules. Molecules that differ less than threefold in size can be separated through the use of highly selective charged membranes and careful optimization of buffer and fluid dynamics.

Current processes often use ion-exchange chromatography, ultrafiltration (UF) and size exclusion chromatography (SEC) for concentration, purification and buffer exchange. HPTFF makes it possible to perform all of these steps in a single-unit operation, thereby reducing production costs. HPTFF can provide high-resolution purification while maintaining the inherent high throughput and high yield characteristics of conventional UF. HPTFF can therefore be used in initial, intermediate and final purification stages.

2. HPTFF operation

2.1. Transmission of proteins through membranes

The separation of proteins by membranes is based on the differential transmission of molecules through the membrane. Typically, one of the species is retained by the membrane while the other species passes through. The transmission of molecules through the membrane is characterized by a parameter called the sieving coefficient as defined in the expression below:

$$S_1 = \frac{C_{p1}}{C_{f1}} \quad (1)$$

where C_{p1} and C_{f1} are the concentrations of species 1 in the permeate and feed, respectively. This parameter is a function of the relative size of the molecule with respect to the size of the pores in the membrane. The size of protein molecules in membrane systems is characterized by a parameter called the effective volume. This represents the effective size of the protein by taking into account the ionic layer associated with the protein in solution. At their isoelectric point, proteins have a net neutral charge. Therefore, no ionic layer is associated with them and their effective volume

is at a minimum. This results in a maximum sieving coefficient. As the pH deviates from the isoelectric point, the net charge on the protein increases, resulting in an increase in the effective volume and the retention of proteins. The response of the effective volume and sieving to the pH is more pronounced at low ionic strength (typically 10 mM). At high ionic strengths, the ionic layer associated with the protein is reduced. This results in high sieving coefficients.

The resolution of proteins is also enhanced by using charged membranes. The sieving of positively charged proteins is dramatically lower in positively charged membranes due to like-charge repulsion. These repulsive interactions prevent the proteins from entering the membranes, which lowers protein transmission and reduces membrane fouling. Selection of a buffer pH where the product protein has positive charge and the impurity proteins have a net neutral charge results in low passage of the product and high passage of the impurities. This enhanced separation is greatest at low ionic strengths, as shown in Fig. 1.

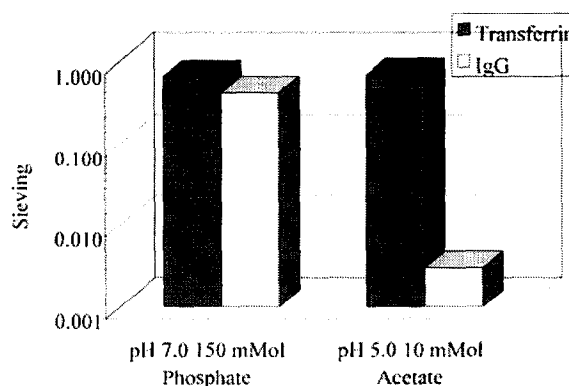


Fig. 1. Effect of pH and ionic strength on the sieving of 6 mg/ml IgG and 1.5 mg/ml transferrin through Biomax™ 100+ membrane in a 50-ml stirred cell. The sieving of IgG can be reduced by two orders of magnitude without reducing impurity passage by optimizing the membrane charge and buffer composition.

3. Case studies

3.1. Fab-BSA separation

Bovine serum albumin (BSA; 68 kD, pI = 4.8) can be removed from an antigen-binding fragment of an antibody (45 kD, pI = 8.6) at pH 5.0 using a positively charged membrane. The reverse separation can be carried out at pH 8.4 using a negatively charged membrane. For more details refer to [6].

3.2. IgG-BSA (Fig. 2)

BSA can be removed from an antibody (160 kD, pI = 8.6) using positively charged membranes. The separation is even more effective than the Fab-BSA separation because of the large size difference between the two proteins.

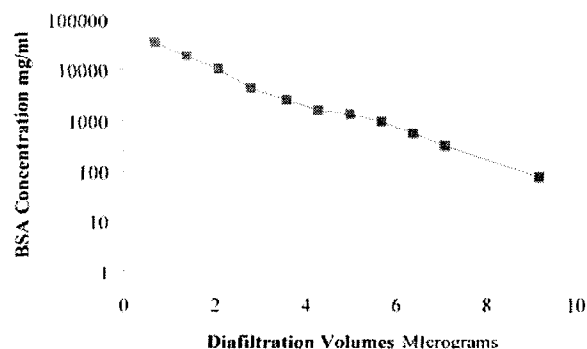


Fig. 2. The removal of BSA from a 17 mg/ml BSA+IgG solution using 10 mM acetate buffer at pH 4.5 with a Ultracell™ 300+ membrane in a Pellicon™ XL 50-cm² device. Elisa purification factors of 10 to 960 were obtained with corresponding IgG yields of >99% in less than 3 h using 6 m² of membrane area per kg of protein. Protein concentration in the retentate as a function of diavolumes (N).

3.3. IgG-transferrin separation (Figs. 3 and 4)

Transferrin (81 kD, pI = 5.0), an impurity that is commonly present from fermentation, can be removed from an antibody (160 kD, pI = 8.6) using positively charged membranes.

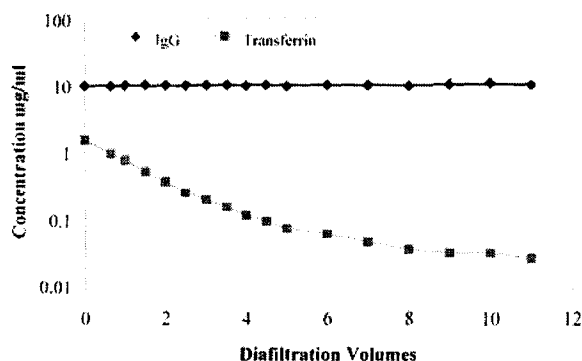


Fig. 3. The removal of transferrin from a 12 mg/ml transferrin+IgG solution using 10 mM acetate buffer at pH 5.0 with a Biomax™ 100+ membrane in a Pellicon™ 20.1 m² device. A280 purification factors of 10–60 were obtained with corresponding IgG yields of >99% in less than 3 h using 17 m² of membrane area per kg of protein. Protein concentration in the retentate as a function of diavolumes (N).

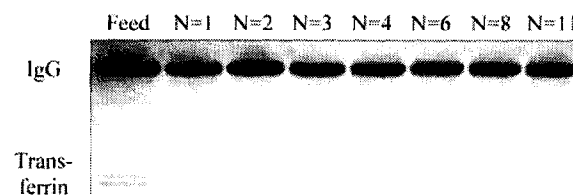


Fig. 4. The removal of transferrin from a 12 mg/ml transferrin+IgG solution using 10 mM acetate buffer at pH 5.0 with a Biomax™ 100+ membrane in a Pellicon™ 20.1 m² device. A280 purification factors of 10–60 were obtained with corresponding IgG yields of >99% in less than 3 h. SDS-PAGE of retentate samples as a function of diavolumes (N).

4. Discussion

HPTFF is an enhanced application of the established technology of process-scale UF and as such can benefit from the well-proven, large-scale membrane and system capabilities already in place. HPTFF exploits multiple phenomena to maximize separation performance. These include the manipulation of solution pH and ionic strength to maximize differences in solute effective volumes, the use of membranes with controlled pore sizes

and surface charge to increase selectivity and throughput and the operation in defined pressure and diafiltration modes to maximize yield and purification factors.

5. Conclusions

HPTFF is a new technology for protein purification that is a two-dimensional separation based on both size and charge. It can be used throughout the purification process and is unique in that it can provide simultaneous concentration, buffer exchange and protein purification. This offers significant potential in process simplification and benefits in reduced capital costs and improved process yields.

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